



Trigonometry · Worksheet 1

Chapter 5: Trigonometry. Answer all questions. Show your working clearly. Total: 196 marks. Time: 4 hours 5 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are ten sections, one for each topic in Chapter 5. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- **Check your calculator mode.** Angles in degrees are marked with $^{\circ}$; otherwise angles are in radians.
- Give non-exact answers to 3 significant figures, or angles to 1 decimal place, unless stated otherwise.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Sine rule, cosine rule and area of a triangle	18	
B Radians: arc length, sector and segment area	18	
C Graphs and solving trigonometric equations	19	
D The identity $\sin^2 \theta + \cos^2 \theta = 1$	21	
E Reciprocal functions: sec, cosec, cot	20	
F Inverse trigonometric functions	17	
G Addition formulae	21	
H Double-angle formulae	19	
I The $R \sin(\theta \pm \alpha)$ / $R \cos(\theta \pm \alpha)$ form	20	
J Small-angle approximations	18	
Roar-level challenge	5	
Total	196	



Section A Sine rule, cosine rule and area of a triangle

18 marks

1. In triangle ABC , $a = 12$ cm, $A = 50^\circ$ and $B = 70^\circ$. Find the length b . [2]

.....

.....

.....

2. In triangle PQR , $PQ = 5$ cm, $PR = 8$ cm and angle $QPR = 60^\circ$. [3]
Find the exact length of QR .

.....

.....

.....

.....

3. Find the area of a triangle with sides 6 cm and 10 cm and included angle 150° . [2]

.....

.....

.....



4. A triangle has sides of length 7 cm, 8 cm and 13 cm.

[4]

(a) Show that the largest angle of the triangle is 120° .

(b) Find the exact area of the triangle.



5. In triangle ABC , $AB = 10$ cm, $AC = 7$ cm and angle $ABC = 35^\circ$.

(a) Show that there are two possible values for angle ACB , and find them both. [3]

.....

.....

.....

.....

.....

(b) Given that angle ACB is obtuse, find the length of BC . [2]

.....

.....

.....

(c) Find the area of triangle ABC in this case. [2]

.....

.....

.....

(Total for Question 5 is 7 marks)

**Section B Radians: arc length, sector and segment area**

18 marks

6. (a) Convert 135° to radians, giving your answer as an exact multiple of π . [2]

(b) Convert $\frac{2\pi}{5}$ radians to degrees.

.....

.....

.....

7. A sector of a circle has radius 5 cm and angle 0.8 radians. [2]
Find the arc length and the area of the sector.

.....

.....

.....

8. A sector of a circle of radius 6 cm has area 24 cm^2 . [3]
Find the angle of the sector, in radians, and the perimeter of the sector.

.....

.....

.....

.....

9. A chord AB of a circle with centre O and radius 8 cm subtends an angle of $\frac{\pi}{3}$ radians at O . [3]
Find the exact area of the minor segment cut off by AB .

.....

.....

.....

.....



10. A sector OAB of a circle has centre O , radius r cm and angle θ radians.
The perimeter of the sector is 20 cm and its area is A cm².

(a) Show that $A = 10r - r^2$.

[3]

.....

.....

.....

.....

.....

(b) Find the maximum possible area of the sector and the value of θ for which this occurs.

[3]

.....

.....

.....

.....

.....

(c) Given that $0 < \theta < 2\pi$, find the range of possible values of r .

[2]

.....

.....

.....

(Total for Question 10 is 8 marks)



Section C Graphs and solving trigonometric equations

19 marks

11. Solve $\cos x = 0.5$ for $0^\circ \leq x \leq 360^\circ$. [2]

.....

.....

.....

12. Solve $\tan x = -\sqrt{3}$ for $0^\circ \leq x \leq 360^\circ$. [3]

.....

.....

.....

13. Solve $\sin 2x = \frac{\sqrt{3}}{2}$ for $0^\circ \leq x \leq 360^\circ$. [4]

.....

.....

.....

.....

.....

14. Solve $3\cos(x - 20^\circ) = 2$ for $0^\circ \leq x < 360^\circ$, giving your answers to 1 decimal place. [3]

.....

.....

.....

.....



15. (a) Sketch the graph of $y = \sin(x + 45^\circ)$ for $0^\circ \leq x \leq 360^\circ$. [2]
Show the coordinates of the points where the graph meets the coordinate axes.

- (b) Solve $2 \sin(x + 45^\circ) = 1$ for $0^\circ \leq x \leq 360^\circ$. [3]

.....

.....

.....

.....

- (c) Hence write down the solutions of $2 \sin(x + 45^\circ) = -1$ for $0^\circ \leq x \leq 360^\circ$. [2]

.....

.....

.....

(Total for Question 15 is 7 marks)



Section D The identity $\sin^2 \theta + \cos^2 \theta = 1$

21 marks

16. Given that $\cos \theta = \frac{5}{13}$ and θ is acute, find the exact values of $\sin \theta$ and $\tan \theta$. [2]

.....

.....

.....

17. Given that $\sin \theta = -\frac{2}{3}$ and $180^\circ < \theta < 270^\circ$, find the exact values of $\cos \theta$ and $\tan \theta$. [3]

.....

.....

.....

.....

18. Prove that $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2 \equiv 2$. [3]

.....

.....

.....

.....



19. Solve $2\cos^2 x + 3\sin x = 3$ for $0^\circ \leq x \leq 360^\circ$.

[5]

.....

.....

.....

.....

.....

.....

.....



20. (a) Show that the equation

[2]

$$3 \sin^2 x + 7 \sin x = \cos^2 x - 4$$

can be written in the form $4 \sin^2 x + 7 \sin x + 3 = 0$.

.....

.....

.....

(b) Hence solve, for $0^\circ \leq x < 360^\circ$,

[4]

$$3 \sin^2 x + 7 \sin x = \cos^2 x - 4,$$

giving your answers to 1 decimal place where appropriate.

.....

.....

.....

.....

.....

.....

(c) Hence find the smallest positive solution of $3 \sin^2 2\theta + 7 \sin 2\theta = \cos^2 2\theta - 4$.

[2]

.....

.....

.....

(Total for Question 20 is 8 marks)



Section E Reciprocal functions: sec, cosec, cot

20 marks

21. Find the exact values of (a) $\sec 45^\circ$, (b) $\cot 30^\circ$. [2]

.....

.....

.....

22. Solve $\operatorname{cosec} x = -2$ for $0^\circ \leq x \leq 360^\circ$. [2]

.....

.....

.....

23. Prove that $\sec \theta - \cos \theta \equiv \sin \theta \tan \theta$. [3]

.....

.....

.....

.....

24. Solve $2 \tan^2 x + \sec x = 1$ for $0^\circ \leq x \leq 360^\circ$, giving your answers to 1 decimal place where appropriate. [5]

.....

.....

.....

.....

.....

.....

.....

.....



25. (a) Show that the equation $\operatorname{cosec}^2 x - \cot x = 3$ can be written as $\cot^2 x - \cot x - 2 = 0$. [2]

.....

.....

.....

(b) Hence solve $\operatorname{cosec}^2 x - \cot x = 3$ for $0 < x < \pi$, giving your answers in radians to 3 significant figures. [4]

.....

.....

.....

.....

.....

.....

(c) Sketch the graph of $y = \sec x$ for $0 \leq x \leq 2\pi$, showing the equations of any asymptotes and the coordinates of any turning points. [2]

(Total for Question 25 is 8 marks)



Section F Inverse trigonometric functions

17 marks

26. Find, in radians, the exact values of (a) $\arccos\left(\frac{\sqrt{2}}{2}\right)$, (b) $\arcsin\left(-\frac{\sqrt{3}}{2}\right)$. [2]

.....

.....

.....

27. Find, in radians, the exact values of (a) $\arctan(-1)$, (b) $\arccos\left(-\frac{\sqrt{3}}{2}\right)$. [2]

.....

.....

.....

28. Find the exact value of $\tan\left(\arcsin\frac{1}{3}\right)$. [3]

.....

.....

.....

.....

29. (a) Solve $\arcsin(2x - 1) = \frac{\pi}{6}$. [3]

(b) State the set of values of x for which $\arcsin(2x - 1)$ is defined.

.....

.....

.....

.....



30. (a) Sketch the graph of $y = \arccos x$, showing the coordinates of the end points and of the point where the graph meets the y -axis. [2]

- (b) By letting $y = \arcsin x$, prove that, for $-1 \leq x \leq 1$, [3]

$$\arcsin x + \arccos x = \frac{\pi}{2}.$$

.....

.....

.....

.....

.....

- (c) Hence solve the equation $\arcsin x = 2\arccos x$. [2]

.....

.....

.....

(Total for Question 30 is 7 marks)



Section G Addition formulae

21 marks

31. Find the exact value of $\cos 15^\circ$.

[2]

.....

.....

.....

32. Show that $\tan 75^\circ = 2 + \sqrt{3}$.

[3]

.....

.....

.....

.....

33. Given that $\sin A = \frac{4}{5}$ and $\cos B = \frac{5}{13}$, where A and B are both acute, find the exact values of

[4]

(a) $\sin(A + B)$,

(b) $\cos(A - B)$.

.....

.....

.....

.....

.....

.....



34. Show that the equation $\sin(x + 30^\circ) = 2\cos x$ can be written as $\tan x = \sqrt{3}$. [4]
Hence solve $\sin(x + 30^\circ) = 2\cos x$ for $0^\circ \leq x < 360^\circ$.

.....

.....

.....

.....

.....

.....

35. (a) Prove that $\cos(\theta + 60^\circ) + \sqrt{3}\sin\theta \equiv \sin(\theta + 30^\circ)$. [3]

.....

.....

.....

.....

.....

- (b) Hence solve $\cos(\theta + 60^\circ) + \sqrt{3}\sin\theta = \frac{1}{2}$ for $0^\circ \leq \theta < 360^\circ$. [3]

.....

.....

.....

.....

- (c) State the maximum value of $\cos(\theta + 60^\circ) + \sqrt{3}\sin\theta$ and the smallest positive value of θ at which it occurs. [2]

.....

.....

.....

(Total for Question 35 is 8 marks)



Section H Double-angle formulae

19 marks

36. Given that $\cos \theta = \frac{1}{3}$, find the exact value of $\cos 2\theta$. [2]

.....

.....

.....

37. Given that $\tan A = 2$ and A is acute, find the exact values of $\tan 2A$ and $\sin 2A$. [3]

.....

.....

.....

.....

38. Prove that [3]

$$\frac{1 - \cos 2x}{\sin 2x} \equiv \tan x.$$

.....

.....

.....

.....

39. Solve $\cos 2x = \sin x$ for $0^\circ \leq x \leq 360^\circ$. [4]

.....

.....

.....

.....

.....

.....



40. (a) By writing $3x$ as $2x + x$, prove that $\sin 3x \equiv 3 \sin x - 4 \sin^3 x$.

[3]

.....

.....

.....

.....

.....

.....

(b) Hence solve $\sin 3x = 2 \sin x$ for $0^\circ < x < 360^\circ$.

[4]

.....

.....

.....

.....

.....

.....

(Total for Question 40 is 7 marks)



Section I The $R\sin(\theta \pm \alpha)$ / $R\cos(\theta \pm \alpha)$ form

20 marks

41. Express $5\sin x + 12\cos x$ in the form $R\sin(x + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. [3]
Give α to 2 decimal places.

.....

.....

.....

.....

42. Express $\sqrt{3}\sin x - \cos x$ in the form $R\sin(x - \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. [3]
Give the exact values of R and α .

.....

.....

.....

.....

43. Express $2\cos \theta - 3\sin \theta$ in the form $R\cos(\theta + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. [3]

.....

.....

.....

.....

44. Using your answer to Question 42, solve $\sqrt{3}\sin x - \cos x = 1$ for $0^\circ \leq x < 360^\circ$. [3]

.....

.....

.....

.....



45. The height, H metres, above the ground of a passenger on a big wheel is modelled by

$$H = 25 - 8 \sin(20t)^\circ - 15 \cos(20t)^\circ,$$

where t is the time in minutes after the ride starts.

(a) Express $8 \sin \theta + 15 \cos \theta$ in the form $R \sin(\theta + \alpha)$, where $R > 0$ and $0^\circ < \alpha < 90^\circ$. Give α to 2 decimal places. [3]

.....

.....

.....

.....

(b) Hence find the maximum height of the passenger above the ground. [2]

.....

.....

.....

(c) Find the time, to the nearest second, at which the passenger first reaches this maximum height. [3]

.....

.....

.....

.....

(Total for Question 45 is 8 marks)



Section J Small-angle approximations

18 marks

46. Use small-angle approximations to estimate the value of $\sin(0.05) + \cos(0.05)$. [2]

.....

.....

.....

47. Show that, for small values of θ in radians, $\cos 2\theta \approx 1 - 2\theta^2$. [2]

.....

.....

.....

48. Given that θ is small and measured in radians, use small-angle approximations to find the approximate value of [3]

$$\frac{1 - \cos 3\theta}{\theta \sin \theta}.$$

.....

.....

.....

.....

49. Show that, when θ is small and measured in radians, [3]

$$\sin 5\theta + \tan 2\theta - \cos 2\theta \approx 2\theta^2 + 7\theta - 1.$$

.....

.....

.....

.....



50. (a) Show that, when θ is small and measured in radians, [3]

$$\frac{1 - \cos 4\theta}{2\theta \sin 3\theta} \approx \frac{4}{3}.$$

.....

.....

.....

.....

.....

- (b) A student is asked to use the approximation $\cos \theta \approx 1 - \frac{\theta^2}{2}$ to estimate $\cos 5^\circ$. [3]

The student writes: $\cos 5^\circ \approx 1 - \frac{5^2}{2} = -11.5$

Explain the error the student has made, and find a correct estimate of $\cos 5^\circ$ to 6 decimal places.

.....

.....

.....

.....

.....

- (c) Find the percentage error in your estimate, compared with the calculator value of $\cos 5^\circ$. [2]

.....

.....

.....

(Total for Question 50 is 8 marks)



ROAR-LEVEL CHALLENGE

5 marks

51. Prove that $\frac{1 - \cos 2\theta}{1 + \cos 2\theta} \equiv \tan^2 \theta$, and hence solve $\frac{1 - \cos 2\theta}{1 + \cos 2\theta} = 3$ for $0^\circ \leq \theta < 180^\circ$. [5]

.....

.....

.....

.....

.....

.....

.....

.....

END OF WORKSHEET

Total: 196 marks